

PALLAVI M

NODE.JS / PYTHON BACKEND DEVELOPER | EXPRESS | FLASK | REST APIs
Bengaluru, India | pallavi.mahesh98@gmail.com | +91-9148404073

PROFESSIONAL SUMMARY

Backend Developer with 4+ years of experience building scalable REST APIs, microservices, and enterprise web applications using Node.js, Express.js, Python, Flask, MySQL, and PostgreSQL. Experienced in backend architecture, authentication systems, Docker deployments, API integrations, and automation scripting for global automotive OEM clients. Strong understanding of microservices, API security, database optimization, and Agile development methodologies. Currently upskilling in AWS cloud and AI-integrated backend applications.

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), TypeScript, Python, SQL

Backend Technologies: Node.js, Express.js, Flask, REST APIs, Microservices, JWT Authentication, Middleware Development

Frontend Technologies: Angular, React, HTML5, CSS3, jQuery

Databases: MySQL, PostgreSQL

Cloud & DevOps: Docker, Jenkins, Nginx, Git, CI/CD Basics

Tools & Platforms: JIRA, Postman, Socket.IO

Concepts: API Security, RBAC, Error Handling, Request Validation, Agile/Scrum

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Tata Elxsi | Bengaluru, India
Nov 2020 – Oct 2024

Project 1: Booking Application – German Automotive OEM

- Developed scalable RESTful APIs and backend services using Node.js and Express.js for enterprise booking workflows.
- Developed and maintained Python Flask APIs and automation scripts for backend integrations, data processing, and workflow automation.
- Built and maintained microservices for reservation management, authentication, and user access control.
- Implemented JWT-based authentication and role-based authorization (RBAC) for secure API access.
- Designed middleware for token validation, centralized error handling, request processing, and API security.

- Optimized MySQL queries and Sequelize ORM operations to improve application performance and data retrieval efficiency.
- Integrated backend APIs with Angular frontend modules and external enterprise systems.
- Containerized backend applications using Docker and supported deployment workflows using Jenkins.
- Collaborated with cross-functional teams and international stakeholders in Agile sprint planning and feature delivery.
- Performed debugging, vulnerability fixes, developer testing, and production issue resolution.

Tech Stack: Node.js, Express, Angular, MySQL, Sequelize, Docker, Jenkins, HTML, CSS, JIRA,JWT

Project 2: CFU Application – US OEM (ADAS Automation)

- Designed **Python automation scripts** to validate ADAS features (Adaptive Cruise Control, Lane Keep Assist, Emergency Braking).
- Created **reusable Python libraries** for test execution and log analysis, increasing automation efficiency by 30%.
- Collaborated with system engineers to identify and reproduce ADAS defects in simulation and real environments.
- Generated detailed test reports and contributed to **overall ADAS software validation**.

Tech Stack: Python

CAREER BREAK

Planned Maternity Break | October 2024 – November 2025

Took a planned career break for maternity and family care. Stayed technically active through backend development practice, REST API projects, and continuous upskilling in Python and Node.js technologies.

EDUCATION

Bachelor of Engineering (Information Science and Engineering)

BMS Institute of Engineering, Bengaluru | CGPA: 8.4 | 2016 – 2020

CERTIFICATIONS

- The Complete Full Stack Development Bootcamp– Udemy
 - The Complete Python Pro Bootcamp – Udemy
-

PROJECT HIGHLIGHTS

- Developed REST APIs and microservices for enterprise automotive applications.
- Implemented Docker-based deployments
- Automated ADAS feature validation using Python scripts, reducing manual effort by 30%.

KEYWORDS FOR ATS

Node.js, Express.js, Python, Flask, Backend Developer, REST APIs, Microservices, JWT Authentication, API Security, MySQL, PostgreSQL, Docker, Jenkins, CI/CD, SQL, Sequelize ORM, Middleware, API Integration, RBAC, Error Handling, Request Validation, Agile, Git, Postman, Backend Architecture, FastAPI, React.js, React Hooks, JSX, Component-Based Architecture.